

Assessment Handbook

dbt Data Engineering Project & Analytics Architecture

Tooling: VS Code + CoPilot
Engine: dbt CLI & DuckDB

EXECUTIVE SUMMARY

- **Defensive Data Pipeline:** Raw data is strictly cleaned and sanitized upfront; invalid data is never surfaced to reporting layers.
- **Performant Revenue Mart:** Built incrementally on append-only completed orders with a monthly grain for maximum execution efficiency.
- **End-to-End Testing & Docs:** Complete test coverage and precise documentation detailing processing decisions, intermediate model design, incremental strategy, and custom business logic.

1. HOW TO RUN THE PROJECT

From the project root, execute the build pipeline:

```
DBT_PROFILES_DIR=. dbt seed && dbt run && dbt test
```

- Uses local DuckDB profile defined in repo.
- Regenerate seed data via: `python scripts/generate_seed_data.py`

2. DATA PROCESSING DECISIONS

orders

Dates & Times: Invalid and future dates are discarded entirely to prevent period distortion, as dates cannot be assumed.

Quality & Cleanliness: Missing, zero, or zero-like amounts, uncompleted orders, and erroneous records are excluded. Deduplication retains only the most recent record per `order_id`.

customers

Uniqueness & Validity: `customer_id` is unique and non-null. Non-unique emails are flagged across affected rows for downstream handling.

products

Validation: Explicitly cast `unit_price` to decimal numeric type.

3. INTERMEDIATE MODEL DESIGN

`int_orders_enriched` joins cleaned orders and customer data while appending key analytical fields. It serves as a stable, reusable foundation for the revenue mart and downstream models.

4. INCREMENTAL STRATEGY

The monthly revenue mart processes incrementally—executing an initial full refresh, followed by processing only new or updated records. This optimizes performance while maintaining complete revenue history by country and month.

5. COMPLEX METRIC CHOICE

Metric: Month-over-Month (MoM) Growth

Compares each country's revenue to its prior month using `LAG()` to surface regional growth patterns and trends for executive reporting.

Limitations to Consider:

- Assumes complete monthly data; missing or delayed records skew calculations.
- Does not account for seasonality, external market factors, or fiscal calendar variations.
- First month per region has no baseline and evaluates to NULL.

6. TESTING APPROACH

Key Tests Implemented

- **Referential Integrity:** Relationship test linking `orders.customer_id` to `customers.id`.
- **Data Quality:** Verified completed orders and enforced valid revenue (`revenue > 0`).
- **Business Rules:** Ensures no products carry a zero or null `unit_price`.

Rationale

- **Validates Integrity:** Guarantees every order links to a valid customer, raising immediate alerts for orphaned records.
- **Enforces Rules:** Strictly filters reporting to fulfilled orders within valid periods with true economic value.
- **Prevents Regressions:** Catches ETL anomalies (duplicate IDs, bad pricing, future dates) before contaminating executive analytics.

N.B. Native dbt built-in tests provide a first line of defense, supplemented by custom SQL scripts in `/tests` run during `dbt test` for comprehensive business coverage.